Intel® Intelligent Systems Centers of Excellence (CoE), HPC Artificial Intelligence and Edge Computing

School of Computing Science and Engineering
Galgotias University

Intel College Excellence Program Project
Synopsis
Phase -2

"Currency Detection using AI/ML"

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BACKGROUND

In this day and age of high-tech machines and emerging technology. Making a forgery rather than earning it lawfully becomes a simple alternative for some ill-intentioned persons.

Because information about the money and how it is created is publicly available on the Internet.

On a daily basis, the average person will not be able to tell the difference between phoney and real notes.

It will become immense hard for a person who is visually challenged and it will be very challenging to tell the difference between a genuine and a fraudulent notes.

PROBLEM IDENTIFICATION

In the modern world, thousands of individuals are conned every day by bogus notes.

A survey by the RBI indicates that this year has seen a 55 percent increase in bogus 2000 bills.

It's not difficult for a normal person to tell a phoney letter from a real one, but for people who are visually impaired, it is more difficult since they cannot distinguish the subtle distinctions. For people who are blind, it might be difficult to determine if a note is real or fake.

A lot of store owners, small vendors, and shopping centres are impacted by the bogus note frauds.

PROPOSED SOLUTION

A practical solution will be a small compact hand-held device which have a camera loaded with the equipped machine learning software which can detect the notes by taking real-time picture of note and Identifying its Accuracy and detecting the currency type through audio and Visual Notification.

The device can be placed on the billing counter of shopping malls, showrooms and small shop vendors.

LITERATURE SURVEY

Real-time image recognition and processing was once thought of as a science fiction concept. Currency Detection has been in application since a while now , such as vending machines, in Banking sectors , but those machines are Big and Bulky which is not portable.

Many researchers have been encouraged to develop robust and efficient automatic currency detection machine [1-5].

Nowadays, it's common to see automatic machines that can recognize banknotes in the dispensers of contemporary goods like candy, soft drink bottles, and bus or train tickets. The primary goal of money identification technology is to recognize and extract both the visible and unseen characteristics of currency notes. Numerous methods have been suggested up until this point to identify the currency note. But the best way is to use the visible features of the note [1].

Size and colour are two examples. However, if the note is soiled or damaged, this method is ineffective. A note's colour characteristics vary significantly if it is filthy. Therefore, it is crucial to use the right algorithm and extract the characteristics from the picture of the currency note to increase the note recognition's accuracy. In order to address the challenges of identifying the numerous facial components in an image or video and uprooting particular facial traits like the eyes, nose, and mouth, the main objective of research over the past 20 years has been to completely automate the face identification system. For accurate face feature identification, feature classifier research has made great progress.

In order to evaluate how much progress has been made from the past to the present and what still needs to be done in the future to implement an improved algorithm that will lead to a successful design and implementation, comparing the performance of existing systems and the new systems built is crucial as we develop new systems with improved algorithms.

We apply here a simple algorithm which works properly. The image of the currency note is captured through a digital camera. The hidden features of the note are highlighted in the ultraviolet light processing on the image is done on that acquired image using concepts like image segmentation, edge information of image and characteristics feature extraction[2-3].

It involves extraction of invisible and visible features of Indian currency notes. This approach consists of different steps like image acquisition, edge detection, gray scale conversion, feature extraction, image segmentation and decision making [4-5]. Acquisition of image is process of creating digital images, from a physical scene. Here, the image is captured by a simple digital camera such that all the features are highlighted. Image is then stored for further processing

In recent years, technological advancements and attention have both been paid to facial recognition.

There are many commercial facial recognition systems available on the market, but their accuracy is unreliable and they shouldn't be relied. In the area of video-based systems with aspects of recognition, tracking, and integration, there had been important efforts. Additionally, fresh datasets are being created for the analysis and assessment of the recognition methods.

It is not an overstatement to say that the majority of the applications currently in use support face recognition and its features, and we are working to expand this capability so that these applications can be utilized in every area of science and study with satisfactory outcomes.

HARDWARE & SOFTWARE REQUIREMENTS

Hardware

- Memory (RAM): 4 GB or more
- Size/Storage: ~Arguable

Repository link: Github Repo

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